EXERCISE 2

General Solutions for the Eight Complex Eigen-data

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Real General Solution • Euler's Formula

STATEMENT

> Write down the general solution of the equation $d\mathbf{x}/dt = A\mathbf{x}$, when the eigenvalues ($\lambda_\pm$) and eigenvectors ($\boldsymbol{\eta}_\pm$) of A are those in the previous exercise.

PHASE 1: SETUP

The eight eigen-data are those of Exercise 1 (listed there). For each, $\boldsymbol{\eta}_+ = \mathbf{u} + i\mathbf{w}$ is the eigenvector of $\lambda_+ = \rho + i\omega$.

PHASE 2: FORMULATION: THE REAL GENERAL SOLUTION

Complex solutions $e^{\lambda_+ t}\boldsymbol{\eta}_+$ and $e^{\lambda_- t}\bar{\boldsymbol{\eta}}_+$ combine into real ones: choose conjugate coefficients, $\mathbf{x} = 2\,\mathrm{Re}\left[Ce^{(\rho+i\omega)t}(\mathbf{u}+i\mathbf{w})\right]$ with $C = \alpha + i\beta$. Expanding $e^{i\omega t} = \cos\omega t + i\sin\omega t$ (Euler) and keeping the real part, with $c_1 = 2\alpha$, $c_2 = -2\beta$ (Topic 4 Eq. 7.3, Robinson Eq. 29.2):

$$\mathbf{x}(t) = e^{\rho t}\Big[\,(c_1\cos\omega t + c_2\sin\omega t)\,\mathbf{u} + (c_2\cos\omega t - c_1\sin\omega t)\,\mathbf{w}\Big]$$

Check of the derivation of the real part: $(a+ib)(\mathbf{u}+i\mathbf{w})$ has real part $a\mathbf{u} - b\mathbf{w}$ because $i\cdot i = -1$; with $a = \alpha\cos\omega t - \beta\sin\omega t$ and $b = \beta\cos\omega t + \alpha\sin\omega t$ this is $\tfrac{1}{2}[(c_1\cos\omega t + c_2\sin\omega t)\mathbf{u} + (c_2\cos\omega t - c_1\sin\omega t)\mathbf{w}]$ after substituting $\alpha = c_1/2$, $\beta = -c_2/2$.

> METHOD OF THE CLASS NOTES [NOTES, P. 10]
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> The notes write the real general solution as $\mathbf{x} = C_1\,\mathrm{Re}\{\mathbf{v}e^{\lambda t}\} + C_2\,\mathrm{Im}\{\mathbf{v}e^{\lambda t}\}$ with $\mathbf{v} = \mathbf{v}_1 + i\mathbf{v}_2$ and $\lambda = \rho + i\omega$. Since $\mathbf{v}e^{\lambda t} = e^{\rho t}(\cos\omega t + i\sin\omega t)(\mathbf{v}_1 + i\mathbf{v}_2)$,
>
> $$\mathrm{Re}\{\mathbf{v}e^{\lambda t}\} = e^{\rho t}(\cos\omega t\,\mathbf{v}_1 - \sin\omega t\,\mathbf{v}_2), \qquad \mathrm{Im}\{\mathbf{v}e^{\lambda t}\} = e^{\rho t}(\sin\omega t\,\mathbf{v}_1 + \cos\omega t\,\mathbf{v}_2),$$
>
> so it is the formula above with $c_1 = C_1$ and $c_2 = C_2$.

PHASE 3: DERIVATION: THE EIGHT SOLUTIONS

(i) $\lambda_\pm = 1 \pm 3i$ with $\boldsymbol{\eta}_\pm = (1,\ 2 \mp i),\ \dot{x} < 0$

$\rho = 1, \omega = 3, \mathbf{u} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \mathbf{w} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$. Substituting in the general formula,

$$\mathbf{x}(t) = e^{1t}\left[(c_1 \cos 3t + c_2 \sin 3t)\begin{pmatrix} 1 \\ 2 \end{pmatrix} + (c_2 \cos 3t - c_1 \sin 3t)\begin{pmatrix} 0 \\ -1 \end{pmatrix}\right]$$

Component by component:

$$x(t) = c_1 e^{t} \cos 3t + c_2 e^{t} \sin 3t, \qquad y(t) = c_1 e^{t} \sin 3t + 2c_1 e^{t} \cos 3t + 2c_2 e^{t} \sin 3t - c_2 e^{t} \cos 3t$$

Check: with $A = \begin{pmatrix} 7 & -3 \\ 15 & -5 \end{pmatrix}$ (Exercise 1), $\dot{\mathbf{x}} - A\mathbf{x} = \mathbf{0}$ identically (verified symbolically).

(ii) $\lambda_\pm = \pm 3i$ with $\boldsymbol{\eta}_\pm = (1 \pm 2i,\ 1 \mp 3i),\ \dot{x} < 0$

$\rho = 0, \omega = 3, \mathbf{u} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \mathbf{w} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$. Substituting in the general formula,

$$\mathbf{x}(t) = e^{0t}\left[(c_1 \cos 3t + c_2 \sin 3t)\begin{pmatrix} 1 \\ 1 \end{pmatrix} + (c_2 \cos 3t - c_1 \sin 3t)\begin{pmatrix} 2 \\ -3 \end{pmatrix}\right]$$

Component by component:

$$x(t) = -2c_1 \sin 3t + c_1 \cos 3t + c_2 \sin 3t + 2c_2 \cos 3t, \qquad y(t) = 3c_1 \sin 3t + c_1 \cos 3t + c_2 \sin 3t - 3c_2 \cos 3t$$

Check: with $A = \begin{pmatrix} -3 & -3 \\ 6 & 3 \end{pmatrix}$ (Exercise 1), $\dot{\mathbf{x}} - A\mathbf{x} = \mathbf{0}$ identically (verified symbolically).

(iii) $\lambda_\pm = -2 \pm i$ with $\boldsymbol{\eta}_\pm = (1 \mp i,\ 3 \pm i),\ \dot{x} > 0$

$\rho = -2, \omega = 1, \mathbf{u} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \mathbf{w} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$. Substituting in the general formula,

$$\mathbf{x}(t) = e^{-2t}\left[(c_1 \cos 1t + c_2 \sin 1t)\begin{pmatrix} 1 \\ 3 \end{pmatrix} + (c_2 \cos 1t - c_1 \sin 1t)\begin{pmatrix} -1 \\ 1 \end{pmatrix}\right]$$

Component by component:

$$x(t) = c_1 e^{-2t}\sin t + c_1 e^{-2t}\cos t + c_2 e^{-2t}\sin t - c_2 e^{-2t}\cos t, \qquad y(t) = -c_1 e^{-2t}\sin t + 3c_1 e^{-2t}\cos t + 3c_2 e^{-2t}\sin t + c_2 e^{-2t}\cos t$$

Check: with $A = \begin{pmatrix} -\frac{5}{2} & \frac{1}{2} \\ -\frac{5}{2} & -\frac{3}{2} \end{pmatrix}$ (Exercise 1), $\dot{\mathbf{x}} - A\mathbf{x} = \mathbf{0}$ identically (verified symbolically).

(iv) $\lambda_{\pm} = -1 \pm i$ with $\boldsymbol{\eta}_{\pm} = (1,\ \pm i),\ \dot{x} > 0$

$\rho = -1, \omega = 1, \mathbf{u} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \mathbf{w} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Substituting in the general formula,

$$\mathbf{x}(t) = e^{-1t}\left[(c_1\cos 1t + c_2\sin 1t)\begin{pmatrix} 1 \\ 0 \end{pmatrix} + (c_2\cos 1t - c_1\sin 1t)\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right]$$

Component by component:

$$x(t) = c_1 e^{-t}\cos t + c_2 e^{-t}\sin t, \qquad y(t) = -c_1 e^{-t}\sin t + c_2 e^{-t}\cos t$$

Check: with $A = \begin{pmatrix} -1 & 1 \\ -1 & -1 \end{pmatrix}$ (Exercise 1), $\dot{\mathbf{x}} - A\mathbf{x} = \mathbf{0}$ identically (verified symbolically).

(v) $\lambda_{\pm} = 2 \pm 2i$ with $\boldsymbol{\eta}_{\pm} = (\pm 3i,\ 5 \mp 4i),\ \dot{x} < 0$

$\rho = 2, \omega = 2, \mathbf{u} = \begin{pmatrix} 0 \\ 5 \end{pmatrix}, \mathbf{w} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$. Substituting in the general formula,

$$\mathbf{x}(t) = e^{2t}\left[(c_1\cos 2t + c_2\sin 2t)\begin{pmatrix} 0 \\ 5 \end{pmatrix} + (c_2\cos 2t - c_1\sin 2t)\begin{pmatrix} 3 \\ -4 \end{pmatrix}\right]$$

Component by component:

$$x(t) = -3c_1 e^{2t}\sin 2t + 3c_2 e^{2t}\cos 2t, \qquad y(t) = 4c_1 e^{2t}\sin 2t + 5c_1 e^{2t}\cos 2t + 5c_2 e^{2t}\sin 2t - 4c_2 e^{2t}\cos 2t$$

Check: with $A = \begin{pmatrix} \frac{2}{5} & -\frac{6}{5} \\ \frac{82}{15} & \frac{18}{5} \end{pmatrix}$ (Exercise 1), $\dot{\mathbf{x}} - A\mathbf{x} = \mathbf{0}$ identically (verified symbolically).

(vi) $\lambda_{\pm} = 5 \pm 3i$ with $\boldsymbol{\eta}_{\pm} = (2 \pm 5i,\ \mp i),\ \dot{x} < 0$

$\rho = 5, \omega = 3, \mathbf{u} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \mathbf{w} = \begin{pmatrix} 5 \\ -1 \end{pmatrix}$. Substituting in the general formula,

$$\mathbf{x}(t) = e^{5t}\left[(c_1\cos 3t + c_2\sin 3t)\begin{pmatrix}2\\0\end{pmatrix} + (c_2\cos 3t - c_1\sin 3t)\begin{pmatrix}5\\-1\end{pmatrix}\right]$$

Component by component:

$$x(t) = -5c_1e^{5t}\sin 3t + 2c_1e^{5t}\cos 3t + 2c_2e^{5t}\sin 3t + 5c_2e^{5t}\cos 3t, \qquad y(t) = c_1e^{5t}\sin 3t - c_2e^{5t}\cos 3t$$

Check: with $A = \begin{pmatrix}-\frac{5}{2} & -\frac{87}{2}\\ \frac{3}{2} & \frac{25}{2}\end{pmatrix}$ (Exercise 1), $\dot{\mathbf{x}} - A\mathbf{x} = \mathbf{0}$ identically (verified symbolically).

(vii) $\lambda_\pm = \pm 7i$ with $\boldsymbol{\eta}_\pm = (1 \pm i,\ -1 \pm 2i)$, $\dot{x} > 0$

$\rho = 0, \omega = 7, \mathbf{u} = \begin{pmatrix}1\\-1\end{pmatrix}, \mathbf{w} = \begin{pmatrix}1\\2\end{pmatrix}$. Substituting in the general formula,

$$\mathbf{x}(t) = e^{0t}\left[(c_1\cos 7t + c_2\sin 7t)\begin{pmatrix}1\\-1\end{pmatrix} + (c_2\cos 7t - c_1\sin 7t)\begin{pmatrix}1\\2\end{pmatrix}\right]$$

Component by component:

$$x(t) = -c_1\sin 7t + c_1\cos 7t + c_2\sin 7t + c_2\cos 7t, \qquad y(t) = -2c_1\sin 7t - c_1\cos 7t - c_2\sin 7t + 2c_2\cos 7t$$

Check: with $A = \begin{pmatrix}-\frac{7}{3} & \frac{14}{3}\\ -\frac{35}{3} & \frac{7}{3}\end{pmatrix}$ (Exercise 1), $\dot{\mathbf{x}} - A\mathbf{x} = \mathbf{0}$ identically (verified symbolically).

(viii) $\lambda_\pm = -13 \pm 17i$ with $\boldsymbol{\eta}_\pm = (\pm 6i - 8,\ 4 \mp 5i)$, $\dot{x} > 0$

$\rho = -13, \omega = 17, \mathbf{u} = \begin{pmatrix}-8\\4\end{pmatrix}, \mathbf{w} = \begin{pmatrix}6\\-5\end{pmatrix}$. Substituting in the general formula,

$$\mathbf{x}(t) = e^{-13t}\left[(c_1\cos 17t + c_2\sin 17t)\begin{pmatrix}-8\\4\end{pmatrix} + (c_2\cos 17t - c_1\sin 17t)\begin{pmatrix}6\\-5\end{pmatrix}\right]$$

Component by component:

$$x(t) = -6c_1e^{-13t}\sin 17t - 8c_1e^{-13t}\cos 17t - 8c_2e^{-13t}\sin 17t + 6c_2e^{-13t}\cos 17t, \qquad y(t) = 5c_1e^{-13t}\sin 17t + 4c_1e^{-13t}\cos 17t + 4c_2e^{-13t}\sin 17t - 5c_2e^{-13t}\cos 17t$$

Check: with $A = \begin{pmatrix}\frac{423}{8} & \frac{425}{4}\\ -\frac{697}{16} & -\frac{631}{8}\end{pmatrix}$ (Exercise 1), $\dot{\mathbf{x}} - A\mathbf{x} = \mathbf{0}$ identically (verified symbolically).

PHASE 4: INTERPRETATION AND CHECKS

- The two constants are real and are the initial coordinates in the $(\mathbf{u}, \mathbf{w})$ basis: $(c_1, c_2)^T = Q^{-1}\mathbf{x}_0$.
- Centres ($\rho = 0$): (ii) and (vii) give purely periodic $\mathbf{x}(t)$ with period $2\pi/3$ and $2\pi/7$; spirals have an envelope $e^{\rho t}$.
- Linear independence: $e^{\rho t}(\cos \omega t\, \mathbf{u} - \sin \omega t\, \mathbf{w})$ and $e^{\rho t}(\sin \omega t\, \mathbf{u} + \cos \omega t\, \mathbf{w})$ have Wronskian $e^{2\rho t} \det[\mathbf{u}\ \mathbf{w}] \neq 0$ since $\mathbf{u}, \mathbf{w}$ are independent (Exercise 4).
- Dimensional check: the exponent ρt and the phase ωt are dimensionless when ρ and ω are rates in s^{-1} and t in s.